

Trial-Termination Early-Warning — live risk dashboard

Three tiers of early-termination risk, monitored continuously. The AI **detects, aggregates, and triages** signals and drafts the alert; the SRC / DSMB / medical monitor, the PM, and the account lead make every call. Rows marked **ADV** come from the [advanced engine](#) — **Optimal Transport (Wasserstein drift)** and **nested Active Inference (free-energy surprise)** — which surfaces risk *continuously and earlier* than a threshold rule. *Values below are illustrative.*

● **PARTICIPANT**

AMBER

DLTs · safety · PK exposure · 17 signals monitored

● S-1042 — candidate Hy's-Law / DILI: ALT 3.4× ULN + total bili 2.1× ULN (DLT window, day 9)

● S-1039 — ΔQTcF +48 ms (watch >30, alert >60 ms)

● Cohort 4 — within-participant AE cluster trending Grade 2 → 3

● ▷ free-energy spike on S-1042 since day 7 — EFE recommends an unscheduled LFT **ADV**

On-device local LLM · zero egress

● **STUDY**

AMBER

Dose-escalation · enrollment · QTLs · 29 signals monitored

● Cohort 4 DLT rate 1/6 — within the 3+3 escalation rule

● Screen-failure 38% vs 25% plan — enrollment KRI breached

● Safety QTL (SAE rate) — within tolerance limit

● ▷ Wasserstein drift of cohort LFT distribution ↑ vs reference barycenter (tail mass moving) **ADV**

Cloud Copilot (aggregate) + on-device drill-down

● **CLIENT / SPONSOR**

AMBER

Commercial early-warning · 19 signals monitored

● AR: \$420K > 60 days past due — payment-aging breach

● Sponsor comms cadence −40% MoM — relationship watch

● 2 scope-reduction change-orders this quarter

● ▷ sponsor-behavior Wasserstein drift ↑ vs baseline — relationship cooling **ADV**

Cloud Copilot · CRM / finance / public filings

Early-warning feed

09:12	● PARTICIPANT	S-1042 candidate Hy's-Law signal — flagged on-device, medical monitor + SRC notified, expedited-reporting clock started
08:55	● PARTICIPANT	▷ Active-inference free-energy on S-1042 elevated since day 7 — EFE recommends an unscheduled LFT (medical-monitor approval pending) ADV
08:40	● STUDY	Screen-failure KRI breached (38% vs 25%) — PM notified; feasibility review proposed
08:05	● STUDY	▷ Wasserstein drift of the cohort-4 LFT distribution rising vs the meta-learned reference — transport map points to the upper tail ADV
Yest 16:05	● CLIENT	AR > 60 days breach (\$420K) — account lead alerted; finance escalation drafted
Yest 11:20	● PARTICIPANT	S-1039 ΔQTcF +48 ms — intensified ECG monitoring flagged to medical monitor
2 d ago	● STUDY	Cohort 4 cleared the 3+3 rule (1/6 DLT) — SRC packet pre-assembled

● termination-level risk ● watch / precursor ● within limits | **On-device** participant-level safety/PK, zero egress **Cloud Copilot** non-sensitive ops/commercial | **ADV** = continuous / anticipatory signal from the [advanced engine](#) (Optimal Transport + Active Inference)

The hybrid architecture

Sensitivity decides the engine. **Participant-level safety, PK, and unblinded data never leave the device** — they run on a local LLM. Non-sensitive operational and commercial signals run on M365 Copilot in the cloud. A scoring layer rolls signals up to a tier RAG status; humans own every decision.

Data sources

EDC / validated safety DB · central lab (LFTs) · ECG · IXRS · CTMS · Smartsheet · finance/AR · CRM · sponsor comms · web (public filings)



Routing by data sensitivity

participant-level safety / PK / unblinded → on-device · non-sensitive ops / commercial → cloud

On-device local LLM

participant safety/PK signals — zero egress, Part-11 audited

M365 Copilot (cloud)

study-ops KRIs & client/commercial signals — under BAA



Advanced engine — Optimal Transport + nested Active Inference

runs inside each tier (on-device for participant-level): Wasserstein drift vs a meta-learned reference, and active-inference free-energy / EFE — continuous, anticipatory, geometry-aware signals on top of the rule-based KRIs. [How it works →](#)



Scoring & aggregation

weighted KRIs / stopping rules roll individual signals up to a Participant / Study / Client RAG status — not an autonomous verdict



Alert & escalation

safety → medical monitor / SRC / DSMB · study → PM · client → account lead — with a human-acknowledgement & disposition log

The rule. The AI flags and prepares; **validated tools and humans decide**. Reported safety numbers come from the validated safety database, never the LLM. The medical monitor / SRC / DSMB make every DLT and safety determination; the PM owns the study; the account lead owns the client signal.

Participant-level signals

All 17 are participant-level safety/PK — processed **on-device**, **zero egress**. The AI flags candidate DLTs and safety patterns **early**, before the formal cohort review, and pre-assembles the SRC packet. It never adjudicates the DLT.

Signal	Detection (what the AI does vs. validated/human)	Engine	Threshold / KRI	Escalation
Candidate DLT in the DLT-evaluation window Safety RED - a confirmed DLT can halt escalation, define the MTD early, or terminate the dose level/cohort	Rule-based: local LLM matches AE grade/term/onset-day/attribution against the prespecified DLT criteria for the assigned dose and flags candidate DLTs in near-real-time; it pre-assembles the case narrative and the running cohort DLT tally for the SRC. The validated safety DB holds the authoritative numbers; the LLM never adjudicates. <i>src: Validated safety DB + EDC AE/CM eCRF; IXRS for dose/cohort assignment; protocol DLT-criteria table</i>	On-device	Any AE meeting the protocol DLT definition (e.g. Grade >=3 non-heme, Grade 4 heme, febrile neutropenia, or any individual-stopping-rule AE) within the DLT window for that dose; design-specific (3+3 / BOIN / mTPI-2 / CRM / accelerated titration)	Immediate flag to medical monitor and SRC/SMC chair ahead of the formal cohort review; pause further dosing/escalation per protocol pending adjudication
Sentinel-participant adverse response (FIH) Safety RED - blocks cohort dosing and can stop the dose level / escalation	Rule-based + model-assisted: local LLM watches sentinel-participant AE/vital/lab streams against the cohort-release criteria and raises a stop-the-cohort flag the moment a sentinel breach appears, before remaining participants are dosed. <i>src: Validated safety DB + EDC; bedside vitals/telemetry; IXRS sentinel flag; protocol sentinel-dosing rule</i>	On-device	Any clinically significant AE / vital-sign / lab change in a sentinel participant within the protocol sentinel window that breaches the cohort-release criteria	Halt dosing of remaining cohort participants; notify PI and medical monitor immediately; convene SRC before proceeding
SAE / death / life-threatening event Safety RED - a related/unexpected SAE or death can stop the dose level or terminate the study	Rule-based: local LLM detects seriousness criteria / Grade 4-5 / fatal outcome on entry, drafts the expedited-reporting case summary and links it to dose, exposure, and concomitant meds for the medical monitor. Authoritative seriousness/causality determination is human + validated PV system. <i>src: Validated safety DB (SAE form) + EDC; PV intake</i>	On-device	Any SAE, death, or life-threatening (Grade 4/5) event regardless of attribution	Immediate medical-monitor and SRC notification; feeds SUSAR/expedited-reporting clock; sponsor + PV informed
Individual AE cluster / severity-grade escalation Safety AMBER - precursor pattern; can convert to a DLT and gate escalation	Model-assisted: local LLM trends each participant's AE grades over time, clusters related MedDRA terms, and surfaces an upward severity trajectory before any single event crosses the DLT line. Triage/explanation only; no decision. <i>src: Validated safety DB + EDC longitudinal AE records (by MedDRA SOC/PT)</i>	On-device	Grade escalation across visits for the same PT, or >=2-3 related AEs clustering in one participant, trending toward a DLT-defining grade	Early heads-up to medical monitor and SRC ahead of the cohort review; flag participant for intensified monitoring
Hy's Law / drug-induced liver injury signal	Rule-based (eDISH logic): local LLM evaluates ALT/AST, total bilirubin, ALP against ULN multiples per participant,	On-device	ALT or AST >3x ULN AND total bilirubin >2x ULN AND ALP <2x ULN with no	Urgent medical-monitor alert; close monitoring / repeat

Safety RED - a Hy's-Law case is a serious DILI signal that can stop dosing for the participant and the program	flags Hy's-Law pattern and component breaches, and assembles the eDISH plot context. Confirmatory repeat-test and adjudication are human + validated lab/safety DB. <i>src: Central-lab LFT feed (ALT, AST, total bilirubin, ALP) + EDC; baseline normals</i>		alternative cause (Hy's Law); component watch-flags at ALT >3x/5x/8x ULN	LFTs per protocol; participant stopping rule and SRC review
QTcF prolongation / ECG abnormality Safety RED at >500 ms / >60 ms delta (potential individual stopping rule and escalation halt); AMBER at watch thresholds	Rule-based per ICH E14: local LLM compares QTcF and delta-QTcF against thresholds, flags new arrhythmia / conduction change, and links the finding to PK exposure timing. Cardiologist/core-lab read is authoritative. <i>src: Central ECG core-lab feed (QTcF, PR, QRS, morphology) + EDC; baseline ECG</i>	On-device	QTcF >500 ms OR delta-QTcF >60 ms from baseline (watch at >450/480 ms or delta >30 ms); new clinically significant arrhythmia or >25% PR/QRS change	Alert medical monitor / cardiac safety reviewer; participant ECG stopping rule; SRC review before next dose
PK exposure outlier vs safety-margin / NOAEL cap PK-PD AMBER/RED - overexposure can gate escalation and trigger early dose-cap or stopping	Statistical: local LLM screens each participant's Cmax/AUC against the prespecified exposure cap and against the cohort distribution (outlier detection), flagging suprathreshold exposure even before an AE manifests. Validated PK software/pharmacometrician owns the formal NCA values. <i>src: Bioanalytical/PK dataset (Cmax, AUC) + EDC dosing; NOAEL-scaled exposure cap from the IB/preclinical package</i>	On-device	Individual Cmax or AUC exceeding the NOAEL-scaled safety-margin cap, or a statistical outlier vs cohort (e.g. > predefined SD/percentile)	Flag to clinical pharmacologist / medical monitor and SRC; informs go/no-go on the next dose level
Unexpected accumulation across MAD dosing PK-PD AMBER - can force regimen change or stop the MAD cohort	Statistical: local LLM tracks each participant's trough/exposure trajectory and accumulation ratio (Racc) against the model-predicted ratio, flagging nonlinear or supra-predicted accumulation that raises steady-state safety risk. <i>src: Bioanalytical/PK trough + profile data across dosing days + EDC; predicted accumulation ratio from SAD/modeling</i>	On-device	Observed accumulation ratio exceeding the predicted Racc by the prespecified margin, or rising troughs without plateau (nonlinear PK)	Alert pharmacometrician / medical monitor; reassess MAD dose/regimen and steady-state exposure cap with SRC
Vital-sign / hemodynamic event Safety AMBER/RED depending on severity and recurrence; can trigger individual stopping and inform cohort review	Rule-based + model-assisted: local LLM compares vitals against protocol alert bands and baseline, detects post-dose hemodynamic shifts time-locked to dosing, and surfaces the trend for the medical monitor. <i>src: EDC vitals + bedside telemetry/orthostatics; validated safety DB</i>	On-device	Vital signs breaching protocol clinically-significant thresholds (e.g. SBP <90 / >180 mmHg, HR <40 / >120 bpm, SpO2 <92%, temp >=38.5C) or a protocol vital-sign stopping rule	Notify PI / medical monitor; apply participant stopping rule; SRC review if recurrent across the cohort

Immunogenicity / infusion (hypersensitivity) reaction Safety RED for anaphylaxis/severe reaction (participant stop + program signal); AMBER for milder/asymptomatic ADA	Rule-based: local LLM flags reactions time-locked to infusion (rash, bronchospasm, hypotension, fever) and links emerging ADA-positive results to clinical events; assembles the case for review. <i>src: EDC AE + infusion-reaction CRF; central-lab ADA/immunogenicity assay; validated safety DB</i>	On-device	Grade >=2 infusion/hypersensitivity reaction, anaphylaxis (any grade), or treatment-emergent ADA positivity associated with a clinical event or PK loss	Immediate medical-monitor alert; premedication/rate-change or stop per protocol; SRC review
Individual protocol stopping-criteria breach Safety RED for the participant; contributes to cohort-level escalation decision	Rule-based: local LLM continuously evaluates each participant's safety data against the full protocol individual-stopping-criteria checklist and fires the moment any single criterion is met, regardless of whether it also qualifies as a DLT. <i>src: EDC + validated safety DB + labs/ECG/vitals; protocol individual-stopping-rule table</i>	On-device	Any breach of the protocol-defined individual stopping/withdrawal criteria (the specific lab/ECG/vital/AE limits listed in the protocol)	Immediate flag to PI / medical monitor; stop that participant's dosing per protocol; document and inform SRC
Participant withdrawal / dropout / early discontinuation Operational AMBER - threatens cohort decision validity and can delay or invalidate the dose-level call	Rule-based: local LLM tracks disposition vs the DLT window and flags participants who become non-DLT-evaluable (incomplete window), which can require replacement to keep the cohort decision valid. <i>src: EDC disposition/EOT records; IXRS; visit-attendance data</i>	On-device	Withdrawal / discontinuation / lost-to-follow-up before completing the DLT-evaluation window, dropping evaluable n below the design requirement	Alert PM and medical monitor; trigger participant replacement per protocol to preserve cohort DLT-evaluability
Dosing non-compliance / PK-affecting deviation Quality AMBER - affects data integrity and can confound the dose-level safety/PK decision	Rule-based + model-assisted: local LLM cross-checks actual vs planned dose, timing, and sampling windows, flags non-compliance and deviations that affect exposure or DLT attribution, and drafts the deviation note. <i>src: EDC dose-administration + deviation log; eCRF; bioanalytical sampling-time records</i>	On-device	Any dosing deviation (wrong dose/time/route, missed/double dose) or sampling-window breach that affects safety or PK evaluability of the participant	Flag to PM / medical monitor and clinical pharmacologist; assess impact on DLT-evaluability and exposure; log deviation
Eligibility violation / enrolled-in-error Quality AMBER/RED - a safety-relevant eligibility breach can require participant withdrawal and signals site/quality risk	Rule-based: local LLM re-checks screening data and post-enrollment findings against the I/E criteria and flags eligibility violations in already-dosed participants (e.g. a screening lab/ECG/medical-history exclusion missed at enrollment). <i>src: EDC screening/eligibility eCRF; central-lab screening labs; medical history</i>	On-device	Confirmed inclusion/exclusion breach in an enrolled-and-dosed participant (e.g. exclusionary lab/ECG/comorbidity/con-med present at dosing)	Immediate notification to PI / medical monitor; protocol-deviation report; assess participant safety and DLT-evaluability impact; SRC awareness
Wasserstein drift vs meta-learned reference ADV OT · drift	Optimal Transport: (sliced-)Wasserstein distance, participant vs reference; the transport map localizes WHICH grades/exposures drive the shift.	On-device	W-distance leaves its control band, trending toward worsening (complements the rule-	Early flag to medical monitor; intensify monitoring

AMBER — continuous precursor, earlier than a threshold rule	Decision-support — flags for review, makes no determination. <i>src: central-lab / ECG / PK streams (on-device)</i>		based DLT/Hy's-Law triggers)	on the implicated measures
Free-energy spike (participant generative model) ADV Active inference AMBER — anticipatory deviation signal	Active inference: variational free energy (negative log model-evidence), precision-weighted per participant. Decision-support only. <i>src: participant safety/PK observations (on-device)</i>	On-device	Free-energy exceeds its rolling band	Flag for medical-monitor review; raise the participant's monitoring precision
EFE-recommended monitoring action (active sampling) ADV Active inference AMBER — proactive monitoring recommendation	Active inference EFE = epistemic (information-gain) + pragmatic (reach the preferred/safe state) value. A SUGGESTION for human approval — never an autonomous clinical or dosing action. <i>src: participant state + protocol-allowed assessments</i>	On-device	EFE-optimal action exceeds a value-of-information threshold	Recommend the data-collection action to the medical monitor for approval

Study-level signals

29 signals: the dose-escalation stopping rules, enrollment/feasibility KRIs, safety QTLs (ICH E6(R3)), data-quality and operational risk. Aggregate, de-identified KRIs run on cloud Copilot; any participant-level drill-down stays on-device.

Signal	Detection (what the AI does vs. validated/human)	Engine	Threshold / KRI	Escalation
Cohort DLT rate breaches dose-escalation stopping/de-escalation rule Safety RED — direct excess-toxicity termination / early-MTD risk; highest-priority safety stop signal	Rule-based (3+3: $\geq 2/3$ or $\geq 2/6$ DLTs) or model-assisted (BOIN/mTPI-2/CRM boundary table; Bayesian early-stop $\Pr(\text{tox at lowest dose} > \text{target}) > 0.95$). The VALIDATED tool computes the rate and boundary; the local LLM only flags that a boundary is approaching/met, drafts the SRC pre-read, and explains contributing cases — it never declares the DLT or the stop. <i>src: Validated dose-escalation tool fed by the safety database + DLT adjudication log; EDC/eCRF DLT window data</i>	On-device	DLT count at dose $\geq$ the design's de-escalation/elimination/stop boundary, or Bayesian stop probability exceeded; also an 'amber' pre-alert at one DLT short of boundary	Immediate escalation to SRC/SMC and medical monitor for the dose-escalation decision (hold escalation / de-escalate / declare MTD / stop); DSMB notified
Aggregate AE/SAE rate vs background/expected Safety RED/AMBER — emerging-safety-profile deterioration; potential safety termination	Statistical trend test (observed-vs-expected, CUSUM/Poisson rate comparison) by the validated safety analytics; local LLM aggregates, narrates the emerging-profile shift and drafts the benefit-risk pre-read. Reported counts always come from the validated safety DB, never the LLM. <i>src: Validated safety database (aggregate); MedDRA-coded AE tables; IB/background reference rates</i>	On-device	Aggregate SAE/AE rate exceeding the protocol/charter expected band or a significant upward CUSUM signal vs prior cohorts/background	Escalate to medical monitor / SRC / DSMB and pharmacovigilance for benefit-risk review
Cohort lab / ECG (QTcF) shift Safety AMBER/RED — organ-toxicity or cardiac-safety signal that can halt escalation	Statistical: cohort mean shift, proportion above CTCAE/regulatory thresholds (e.g., QTcF $> 450/480/500$ ms, $\Delta\text{QTcF} > 60$ ms), Hy's-Law-pattern aggregation. Validated lab/ECG analytics compute; local LLM clusters and explains the dose-response trend. <i>src: Central-lab feed + ECG core-lab (validated), aggregated by cohort/dose</i>	On-device	Cohort proportion or mean exceeding pre-defined organ-safety/QTcF thresholds, or a dose-response trend across cohorts	Escalate to medical monitor / SRC / DSMB; cardiac-safety or hepatic-safety review
Safety Quality Tolerance Limit (QTL) breach Quality AMBER — systemic data-reliability/safety-quality	Rule-based threshold crossing per the ICH E6(R3) RBQM plan; cloud Copilot flags the breach, drafts the documented root-cause/CAPA narrative and routes it. QTL is a quality trigger, not an auto-stop.	Cloud Copilot	QTL parameter outside its pre-specified acceptable range (with secondary/warning limit for early alert)	Escalate to PM / quality lead / sponsor for root-cause assessment and possible protocol/process

risk; reputational + reliability termination risk	<i>src: Aggregate EDC + safety DB QTL metrics; RBQM/CSM platform (de-identified)</i>			change; document per E6(R3)
Enrollment lag vs plan (recruitment KRI) Enrollment AMBER — feasibility & timeline termination risk	Rule-based actual-vs-planned variance + trend/forecast; cloud Copilot computes the gap, projects completion date and drafts the recruitment-risk brief. <i>src: CTMS / IXRS-IWRS enrollment feed; recruitment plan (de-identified counts)</i>	Cloud Copilot	Cumulative enrollment < X% of plan for ≥N weeks, or projected last-participant-in slips past the critical-path milestone	Escalate to PM and sponsor; feasibility/site-add or de-scope discussion
Screen-failure / screen-to-enrollment ratio breach Enrollment AMBER — feasibility / cost-overrun termination risk	Rule-based ratio vs assumption + trend; cloud Copilot trends the ratio and identifies driving eligibility criteria from de-identified screen-fail reasons. <i>src: CTMS / EDC screening logs; IXRS (de-identified aggregate ratios)</i>	Cloud Copilot	Screen-failure rate above the protocol-assumed band, or worsening trend across cohorts	Escalate to PM / medical monitor; consider protocol amendment to eligibility
Futility / low probability of meeting the primary objective Operational RED/AMBER — futility / program non-viability termination risk	Model-assisted: dose-escalation model projection / conditional-power / predictive-probability from the validated tool; local LLM narrates the futility scenario where participant-level exposure/safety is involved. <i>src: Validated dose-escalation model + PK exposure summaries; SAP interim analysis (de-identified projections)</i>	On-device	Predicted primary-objective success below the pre-specified futility bound, or MTD/exposure trajectory off-target	Escalate to DSMB/SRC and sponsor for futility/continue decision
Query rate & query-aging KRI Quality GREEN/AMBER — data-quality & lock-timeline risk	Rule-based KRI thresholds + aging buckets; cloud Copilot trends rate/aging by site and drafts the data-management escalation. <i>src: EDC query management (de-identified aggregate metrics)</i>	Cloud Copilot	Query rate above KRI limit or % queries open >X days above threshold / rising trend	Escalate to data management lead / PM; site follow-up
Missing-data & overdue-CRF trend Quality AMBER — reliability & lock-timeline risk	Rule-based completeness KRI + trend; cloud Copilot flags critical-variable gaps and drafts the completeness alert. <i>src: EDC completeness metrics (de-identified aggregate)</i>	Cloud Copilot	Missing critical-data % or data-entry lag above KRI limit / upward trend; ties to a safety QTL when critical	Escalate to data management / PM; if safety-critical, medical monitor
Protocol-deviation rate & trend Quality AMBER/RED — GCP-conduct & reliability termination risk	Rule-based KRI + trend with severity weighting; cloud Copilot trends and clusters deviation types, drafts the GCP-risk brief. Important-deviation clustering can feed a QTL. <i>src: CTMS / EDC deviation log (de-identified aggregate; severity-coded)</i>	Cloud Copilot	Critical/important deviation rate above KRI limit or rising trend, or a single critical safety deviation	Escalate to PM / quality / medical monitor; CAPA, possible for-cause review
IXRS / randomization & drug-assignment	Rule-based anomaly detection on transactions; cloud Copilot flags	On-device	Any unblinding/wrong-dose dispensing event, or	Escalate to PM / medical monitor

Integrity Operational AMBER/RED — integrity & safety termination risk	aggregate error trend, local LLM handles any participant-level/unblinding drill-down. <i>src: IXRS/IWRS transaction logs + drug-accountability (de-identified aggregate; participant-level for the specific event)</i>		IXRS error rate above threshold	(safety) and quality; possible for-cause
Central statistical-monitoring data-integrity anomaly Quality AMBER/RED — data-integrity / fraud termination & reliability risk	Statistical (validated CSM): unsupervised multivariate tests, digit-preference & variance tests, site-outlier profiling. The CSM tool computes the signal; cloud Copilot narrates and prioritizes the flagged sites, drafts the central-monitoring report — it does not adjudicate fraud. <i>src: RBQM/CSM platform over all EDC data (validated statistical engine; de-identified aggregate site profiles)</i>	Cloud Copilot	CSM anomaly score / p-value past the platform threshold for a site or variable cluster; under-reporting flag vs study mean	Escalate to quality / biostatistics / PM; targeted SDV or for-cause audit
Site / operational performance outlier Operational AMBER — site-conduct & reliability risk	Model-assisted site-profiling / weighted composite KRI ranking; cloud Copilot ranks sites, explains the driving metrics, drafts the site-management plan. <i>src: CTMS + EDC + IXRS composite site scorecard (de-identified aggregate)</i>	Cloud Copilot	Composite site-risk score above threshold or KRI breaches on ≥N metrics	Escalate to PM / CRA lead; targeted monitoring, retraining, or site closure
Timeline slip & budget-burn vs milestones Financial AMBER — timeline & cost termination/de-scope risk	Rule-based earned-value / schedule-variance + forecast; cloud Copilot computes schedule & cost variance, projects end-date and burn, drafts the program-risk brief. <i>src: CTMS / Smartsheet project plan + finance (de-identified milestone & cost metrics)</i>	Cloud Copilot	Critical milestone forecast past baseline by >X, or cost-to-progress variance beyond threshold	Escalate to PM / program lead and sponsor; replan / change-order discussion
SUSAR / expedited-reporting clock breach Pharmacovigilance / regulatory timeliness Critical	Clock countdown vs 7/15-day deadline; backlog of un-reconciled serious events pending causality/expectedness assessment <i>src: Safety database / PV system (validated); investigator awareness timestamps</i>	On-device	Alert at T-minus-X days before 7- or 15-day deadline; any breach = critical	Draft expedited-report status alert to PV/medical monitor and regulatory; never auto-submit or auto-assess causality
Clinical hold / partial hold / IND safety-report regulatory action Regulatory standing Critical	Detect hold notice receipt; or aggregate-serious-event pattern crossing IND-safety-report logic <i>src: Regulatory correspondence inbox, FDA/agency portals, sponsor regulatory team; safety aggregate from validated DB</i>	On-device	Any hold/partial-hold = critical; aggregate pattern = warning to medical monitor	Draft regulatory-action alert; flag dosing-pause implications; humans own filing and hold response
DSMB / IDMC / SRC recommendation-to-	Recommendation = stop/pause/modify/hold-	On-device	Any stop/pause recommendation = critical;	Draft governance-recommendation alert

stop or pause Safety-governance decision Critical	escalation; or sponsor non-adoption of a safety recommendation <i>src: DSMB/IDMC charter outputs, SRC minutes, recommendation letters</i>		modify/no-escalate = warning	and route to PM/sponsor; committee owns the recommendation, sponsor owns the decision
DSMB/SRC meeting-cadence & dose-escalation-decision data-package readiness Safety-governance operations Warning	Days-to-meeting vs data-readiness; missed/no-quorum meeting; package incompleteness at the gate <i>src: Committee charter cadence, meeting scheduler, data-cleaning/lock status of the cohort package</i>	On-device	Package not ready by T-minus-N business days = warning; missed gating meeting = critical (blocks escalation)	Draft readiness/cadence alert to data management and PM; humans schedule and decide
Aggregate benefit-risk / emerging-safety-profile shift (DSUR-style) Aggregate safety surveillance Warning	Disproportionality / trend on cumulative MedDRA-coded events vs prior cohorts and reference; central-tendency drift <i>src: Validated safety database, central labs/ECG, cumulative exposure</i>	On-device	Statistically/clinically meaningful upward shift in a term cluster or organ system	Draft emerging-profile summary for medical monitor / aggregate-safety review; never re-characterize benefit-risk autonomously
Bioanalytical assay failure / sample-integrity loss affecting PK Bioanalytical / data integrity Warning	Run-rejection rate, QC/calibration failures, ISR pass-rate breach, missing/excursion samples per cohort <i>src: Bioanalytical lab LIMS, sample-accountability/courier logs, QC/ISR reports</i>	On-device	ISR pass-rate < acceptance (e.g., regulatory ISR criteria), or assay-run rejection / sample-loss above tolerance for a gating cohort	Draft PK-integrity alert to clinical pharmacology/PK scientist; flag escalation-decision risk; humans adjudicate data usability
Investigational product supply: stock-out / expiry / shipment & cold-chain failure Supply chain / IP accountability Warning	Projected days-of-supply vs enrollment plan; expiry runway; excursion quarantine; release delay <i>src: IXRS/IRT supply module, depot inventory, QP/QA release status, temperature-monitoring logs</i>	Cloud Copilot	Projected supply < lead time to next cohort; remaining shelf-life < dosing window; any cold-chain quarantine of gating stock	Draft supply-risk alert to clinical supply/PM; humans manage resupply and QP release
Specialty-vendor / lab / ECG-core / IXRS outage or performance failure Vendor operational integrity Warning	Turnaround-time SLA breach, missed/late data feeds, system-uptime drop, reconciliation mismatch count <i>src: Vendor SLA dashboards, data-transfer/reconciliation logs, helpdesk/incident tickets</i>	Cloud Copilot	SLA breach on a gating deliverable (e.g., safety labs/ECG before dosing review), or feed gap > tolerance	Draft vendor-performance alert to vendor manager/PM; humans manage escalation/contingency
Informed-consent invalidity / re-consent trigger breach Compliance / human-participant protection Critical	ICF version mismatch vs current approved; re-consent not completed within window after new safety info; IRB continuing-review lapse <i>src: IRB/EC approval records, ICF version control, participant consent log, amendment tracker</i>	On-device	Any dosed participant on a superseded ICF, or re-consent overdue past defined window = critical compliance flag	Draft consent/re-consent alert to clinical operations/regulatory; humans manage IRB and re-consent

Protocol-amendment churn / substantial-amendment frequency Protocol stability Warning	Amendments per unit time vs benchmark; share that are substantial/safety-driven; shortening interval <i>src: Protocol version control, amendment log, IRB/EC submission tracker</i>	Cloud Copilot	Amendment rate or substantial-amendment count above benchmark for study phase/duration	Draft amendment-churn trend to PM/sponsor; humans assess design and program implications
Site for-cause / sponsor / regulatory inspection findings (data integrity / GCP) Quality / inspection Critical	Critical/major finding count, for-cause trigger, fraud-pattern flags from CSM, repeat/un-resolved CAPA <i>src: Audit/inspection reports, monitoring CAPA log, central statistical monitoring outputs, regulatory inspection notices</i>	On-device	Any critical finding or for-cause/fraud signal; major-finding accumulation above tolerance	Draft inspection/audit-risk alert to QA/PM/sponsor; humans own CAPA and regulatory response
Cohort distributional shift across escalating doses (Wasserstein) ADV OT · drift AMBER — continuous dose-escalation safety drift	Optimal Transport on aggregate distributions; the transport plan shows which severity grades / exposure ranges drive the shift. <i>src: aggregate (de-identified) cohort safety/PK</i>	On-device	W-distance trend exceeds the cohort-to-cohort control band	Surface to the SRC ahead of the escalation review
Study-level free-energy drift (model evidence) ADV Active inference AMBER — futility / safety / feasibility precursor	Active inference at the study level; the QTLs encode the 'preferred' states (prior preferences C). Free energy over aggregate states. <i>src: aggregate study KRIs</i>	Cloud Copilot	Sustained free-energy rise above band	PM review; SRC if safety-driven
EFE-guided risk-based monitoring allocation ADV Active inference GREEN — optimization of effort, not a risk itself	Active inference precision/EFE ranks monitoring targets; decision-support for the monitoring plan. <i>src: site/cohort KRIs + uncertainty</i>	Cloud Copilot	EFE ranking of targets	Feed the risk-based monitoring plan; CRA tasking (human-approved)
Site data-distribution anomaly (Optimal Transport) ADV OT · quality AMBER — data-integrity early warning	OT distance, site vs pool; a large distance = outlier site. Decision-support; triggers for-cause review. <i>src: per-site aggregate (de-identified) data</i>	Cloud Copilot	Site W-distance beyond the pool's tolerance	Flag for central-monitoring / for-cause review

Client / sponsor-level signals

19 **commercial** early-warnings that a sponsor may cancel, de-scope, or not renew — non-sensitive business data on cloud Copilot. This is **signal & triage for the account team**, not a verdict; the relationship owner decides.

Signal	Detection (what the AI does vs. validated/human)	Engine	Threshold / KRI	Escalation
AR / payment aging breach Financial AMBER→RED: hard early signal of de-scope/non-payment/termination risk; oldest, most reliable financial tell	Rule-based aging buckets computed by the validated finance system; Copilot AGGREGATES across invoices, correlates with disputes/cadence, and DRAFTS the AR-risk brief. Reported dollar figures come from ERP, never the LLM. <i>src: Finance/AR sub-ledger + ERP (system of record), contract payment terms from CLM/CTMS</i>	Cloud Copilot	Any material invoice >60 days past Net-30/45 terms, OR total past-due trending up 2+ consecutive periods, OR a single invoice in dispute >\$X	Account/BD lead + PM + finance/AR controller; escalate to relationship-governance call
Change-order / scope-reduction / hold trend Commercial AMBER→RED: directional de-scope often precedes full cancellation	Rule-based on CO direction/value + status aging; Copilot classifies CO sentiment (reduce vs. expand), flags stall/hold patterns, and explains. PM validates. <i>src: CTMS / Smartsheet / CLM change-order log, contract scope, project plan</i>	Cloud Copilot	≥2 reduction/hold COs in a quarter, OR a CO unsigned/stalled >30 days, OR cumulative scope down vs. baseline contract value	Account lead + PM; trigger scope/relationship review with sponsor
Cancellation-rate / backlog-erosion spike Commercial AMBER→RED: leading indicator of revenue loss; feeds book-to-bill / NDR	Statistical vs. benchmark — Copilot compares realized erosion to the ~4%-of-prior-quarter-backlog cancellation baseline and flags excess; finance validates the numbers. <i>src: Backlog/financial-planning system, CTMS revenue forecast, contract value</i>	Cloud Copilot	Quarterly backlog burn on this program exceeds ~4% baseline cancellation rate, OR forecast revenue revised down >X% unrelated to milestones	Account lead + portfolio/finance leadership
Communication cadence drop Commercial AMBER: classic B2B churn precursor; soft but early	Statistical anomaly on cadence/latency vs. the relationship's own baseline. Copilot reads METADATA/cadence (not sensitive bodies) and de-identified summaries; flags the drop. Account lead interprets. <i>src: Sponsor comms metadata (email/calendar), CRM activity log, meeting minutes</i>	On-device	Meeting/email cadence down >40–50% vs. trailing baseline, OR median response latency doubles, OR a governance meeting cancelled/not rescheduled	Account/BD lead; proactive re-engagement / health check
Sentiment decline in sponsor comms Commercial AMBER: corroborating signal, weak alone — pair with cadence/disputes	Model-assisted NLP sentiment/topic on de-identified comms; Copilot trends sentiment and surfaces representative quotes, with confidentiality guardrails. Human reads context — sentiment is noisy, triage-only.	Cloud Copilot	Sentiment trend negative for 2+ consecutive periods, OR ≥N negative/escalation-flagged messages in a window	Account lead; loop in PM/relationship sponsor

	src: Email/meeting-minutes text (de-identified), CRM notes, escalation log			
Loss of internal champion Commercial AMBER→RED: champion loss frequently precedes vendor switch	Rule-based on stakeholder activity + public role-change detection; Copilot flags champion silence or exit and maps remaining relationship coverage. Account lead confirms. <i>src: CRM stakeholder map, comms metadata, LinkedIn/press (public), org changes</i>	Cloud Copilot	Champion comms cease >X weeks, OR public departure/role change, OR new unknown decision-maker inserted	Account/BD lead; rebuild multi-threaded relationship
Milestone dispute / escalation flag Commercial AMBER→RED: disputes are both a cause and a pretext for termination	Rule-based on escalation/dispute events; Copilot clusters recurring complaint themes and links to invoice/CO status. PM/quality validate the facts. <i>src: CTMS milestone log, escalation/issue tracker, meeting minutes, contract</i>	Cloud Copilot	≥1 formal escalation, OR ≥2 milestone/acceptance disputes in a quarter, OR sponsor micromanagement spike (a trust tell)	PM + account lead; quality/governance review
Sponsor pipeline / portfolio reprioritization Commercial RED: #1 non-financial cause of early-phase cancellation	Model-assisted web/filing monitoring; Copilot detects pipeline-graphic changes, 'strategic review', 'focusing resources' language, and trial-status flips, then summarizes asset-specific risk. BD lead corroborates. <i>src: Public 8-K/press/earnings calls, clinicaltrials.gov status, conference abstracts, CRM intel</i>	Cloud Copilot	Public deprioritization/shelving language for this asset/TA, OR pipeline removal, OR clinicaltrials.gov status → suspended/withdrawn	Account/BD lead + portfolio leadership; renewal-defense plan
Sponsor financial health / runway Financial RED: insolvency-driven termination ignores trial performance	Model-assisted filing analysis; Copilot extracts cash, burn, and quarters-of-runway, flags going-concern language, and trends. Figures cited to the filing, not invented. Finance/BD validate. <i>src: Public 10-Q/10-K, 8-K, press, IR decks, analyst notes; (private) news of raises/runway</i>	Cloud Copilot	Cash runway < ~12 months (or below next-milestone date), OR going-concern qualification, OR failed/withdrawn raise, OR debt covenant breach	Account/BD lead + finance + leadership; assess pre-payment/credit terms
Layoffs / restructuring signal Financial AMBER→RED: strong corroborator of runway/reprioritization risk	Rule + model-assisted news monitoring; Copilot detects layoff/restructuring events for the sponsor and links to affected programs. BD lead confirms relevance to our asset. <i>src: Public press / layoff trackers / WARN notices / 8-K / LinkedIn (public)</i>	Cloud Copilot	Any announced layoff/restructuring/program-cut at the sponsor, especially clinical-ops or our TA	Account/BD lead; proactive status check on our program
Sponsor M&A / acquisition / reverse-merger Commercial RED: step-change; acquirers routinely cut	Model-assisted news/filing monitoring; Copilot flags rumor→announcement→close and cross-references OUR change-of-control / termination-for-convenience terms. Legal/BD	Cloud Copilot	M&A rumor/announcement/definitive agreement involving the sponsor; change-of-control clause present in our MSA	Account/BD lead + legal + leadership; engage acquirer early

acquired Phase 1 programs	validate. <i>src: Public M&A news/8-K/definitive agreements, rumor trackers, CRM intel, our contract terms (CLM)</i>			
Competitive / adjacent-program readout Commercial AMBER→RED: can trigger a 'why continue?' reprioritization	Model-assisted web monitoring keyed to target/MoA/indication; Copilot summarizes the competitive event and reasons about spillover risk to our asset. Scientific/BD leads judge relevance. <i>src: Public readouts, ASCO/AACR/JPM abstracts, FDA actions, press, clinicaltrials.gov</i>	Cloud Copilot	Competitor approval or strong readout in same indication, OR adjacent/same-MoA program safety or efficacy failure	Account/BD lead + scientific lead; sponsor strategy check-in
Contract end-date / renewal & option-exercise window Commercial AMBER: cleanest, earliest deterministic lead time — act before it goes red	Rule-based deterministic countdown; Copilot tracks days-to-decision and un-exercised options and DRAFTS the renewal-defense brief. PM/account lead own the play. <i>src: Contract/CLM metadata, CTMS milestones, option schedule, notice-period clocks</i>	Cloud Copilot	<90 days to contract end / renewal / option-exercise date, OR option past its trigger milestone un-exercised	Account/BD lead + PM; launch renewal/extension campaign
Relationship governance / KOL-strategy pivot flags Commercial AMBER→RED: governance + leadership pivots reshape vendor decisions	Rule + model-assisted; Copilot flags governance-meeting anomalies, detects new sponsor leadership / strategy-pivot language, and surfaces RFP-shopping tells. Account lead interprets intent. <i>src: Governance meeting minutes/cadence, CRM intel, public leadership-change news, RFP/procurement signals</i>	Cloud Copilot	Cancelled/skipped governance meeting, OR new CMO/CSO/dev-strategy change, OR detected competitive RFP / second-CRO sourcing	Account/BD lead + relationship sponsor; executive re-engagement
Same-compound / cross-study & class safety signal Cross-program safety intelligence Critical	New same-compound/class serious finding, regulatory safety communication, competitor program safety halt for the target/class <i>src: FAERS/FDA safety communications, ClinicalTrials.gov, literature, sponsor cross-study safety, regulatory bulletins</i>	On-device	Any credible same-compound or mechanism-class serious safety event	Draft cross-study/class safety brief to medical monitor/sponsor; humans assess applicability and action
Competitor approval / pivotal readout (positive OR negative) reshaping program value Competitive / market intelligence Warning	Competitor approval/designation event; pivotal-trial readout in same indication/target/MoA <i>src: FDA/EMA approval trackers, competitor press/IR, ClinicalTrials.gov, conference/abstract readouts, analyst coverage</i>	Cloud Copilot	Any first-in-class approval or major pivotal readout in the same indication/target	Draft competitive-event brief to PM/BD/sponsor relationship lead; humans assess portfolio impact

FX / contract-currency & funding-tranche exposure Financial / contractual Warning	FX move beyond contract tolerance band; missed/at-risk funding tranche; going-concern flag; covenant breach <i>src: FX rates vs contract baseline, sponsor SEC/financial filings, funding/milestone tranche schedule, going-concern disclosures</i>	Cloud Copilot	FX deviation beyond agreed band; any funding tranche slip or going-concern disclosure	Draft financial-exposure alert to finance/PM/account lead; humans manage contract and currency hedging
Sponsor-behavior drift (Wasserstein) ADV OT · drift AMBER — commercial early warning	Optimal Transport on engagement features; a rising distance = the relationship cooling. <i>src: CRM / finance-AR / comms metadata (non-sensitive)</i>	Cloud Copilot	W-distance beyond the baseline band	Account / BD lead review
Program-level free-energy (commercial surprise) ADV Active inference AMBER — program/contract-risk precursor	Active inference at the client level; decision-support for the account team. <i>src: program milestones / comms / AR</i>	Cloud Copilot	Sustained free-energy rise	Account lead + leadership review

Detection, scoring & escalation

How individual signals become a tier risk level and an alert — and the platform pieces that run it.

Signal	Detection (what the AI does vs. validated/human)	Engine	Threshold / KRI	Escalation
On-device local LLM backend (sensitive participant-level engine) Platform / AI engine n/a	Local open-weight LLM runs RAG/summarization over participant records on-device; zero network egress; the model triages and EXPLAINS, validated tools compute the reportable safety numbers and the SRC/medical monitor makes the call <i>src: EDC, validated safety/PV database, central-lab, ECG core-lab, IXRS (participant-level)</i>	On-device	n/a — capability; sensitivity classification 'participant-level/unblinded' forces local-only processing	Drafts participant-level early-warning narratives for the medical monitor / SRC; no data leaves the device
Cloud M365 Copilot backend (non-sensitive ops/commercial engine) Platform / AI engine n/a	Cloud Copilot summarizes and triages non-sensitive KRIs and commercial signals; drafts alerts for PM and account/BD lead; validated tools/CSM compute the metrics <i>src: CTMS, Smartsheet, finance/AR, CRM, sponsor-comms, web (public filings) — de-identified/aggregate only</i>	Cloud Copilot	n/a — capability; only data classified non-sensitive/de-identified is permitted to the cloud	Drafts study-ops and commercial early-warning summaries for PM and account/BD lead
Sensitivity-based routing gateway Platform / Governance routing n/a	Rule-based gateway at the connector layer tags each record's sensitivity class and routes accordingly before any model sees it; routing decisions are logged <i>src: Data-classification metadata on every connector</i>	Validated + human	n/a — deterministic policy rule (sensitivity class = local-only vs cloud-eligible)	Blocks any participant-level/unblinded payload from reaching the cloud; logs the routing decision to the Part-11 audit trail
Validated data connectors (read-only ingestion) Platform / Data connectors n/a	Read-only, validated, time-stamped pulls with each feed tagged by sensitivity class; reportable numbers sourced from validated systems, not the LLM <i>src: EDC, validated safety DB, IXRS/RTSM, central-lab, ECG core-lab, CTMS, Smartsheet, finance/AR, CRM, sponsor-comms, web (public filings)</i>	Validated + human	n/a — capability; connector validation and ALCOA++ lineage required	Feeds the scoring/aggregation layer and the appropriate (local/cloud) engine
3-tier RAG dashboard view (Participant / Study / Client) Platform / Visualization n/a	Renders weighted-KRI roll-up as RAG tiles; sparklines show signal trajectory over time; drill-down routes participant-level detail back through the on-device engine <i>src: Scoring/aggregation layer (all connectors)</i>	On-device	n/a — display layer; RAG bands inherited from the scoring layer's KRI/QTL thresholds	Gives the medical monitor/SRC, PM, and account lead a single triage surface to review and disposition
Scoring & aggregation layer (weighted-KRI roll-up)	Statistical/rule-based weighted scoring (config-controlled weights and thresholds); explainable — every score links back to its contributing signals; it is	Validated + human	n/a — capability; per-tier KRI weights and QTL/acceptable-	Sets tier RAG status; any red/amber requires

Platform / Scoring n/a	triage support, NOT an autonomous verdict <i>src: All signal feeds (participant, study, client)</i>		range' bands are pre-specified and version-controlled	human review and disposition
Alerting & escalation routing Platform / Alerting n/a	Rule-based routing on signal category and severity; AI drafts the alert text and rationale, the human owner triages and decides <i>src: Scoring/aggregation layer + signal metadata</i>	Validated + human	n/a — capability; escalation tier triggered by the signal's RAG band and category	Delivers the right alert to the right decision-maker with SLA tracking
Human acknowledgement & disposition log Platform / Governance n/a	Each alert requires an attributable, time-stamped human disposition with rationale before close; enforces 'AI flags, human decides' <i>src: Alert workflow</i>	Validated + human	n/a — capability; no red/amber alert closes without a recorded human disposition	Creates the accountable record that a human owned the call
21 CFR Part 11 audit trail (ALCOA++) Platform / Compliance n/a	Rule-based logging with e-signatures and version control; ALCOA++ data lineage; tamper-evident <i>src: All platform actions and engine outputs</i>	Validated + human	n/a — capability; Part-11 / ALCOA++ compliance required for all entries	Provides the regulatory-inspectable record of detection-to-disposition
Tiered refresh cadence Platform / Operations n/a	Near-real-time / event-driven for participant safety (EDC, safety DB, IXRS, labs, ECG); daily for study-ops KRIs (CTMS, queries, deviations, Smartsheet); weekly for client/commercial (finance-AR, CRM, sponsor comms, public filings) <i>src: Per-connector schedules / event triggers</i>	Validated + human	n/a — capability; cadence set per data domain	Ensures safety signals surface fast while commercial signals refresh on a fit-for-purpose cycle
Explainability & source-citation layer Platform / Trust n/a	RAG with mandatory source citation; the model explains WHY a signal fired and links to source; reviewer confirms against the validated safety DB <i>src: All connectors + scoring layer</i>	On-device	n/a — capability; no uncited safety claim is shown as fact	Lets the human owner trust-but-verify before dispositioning
Role-based access & blinding controls Platform / Security n/a	Rule-based RBAC and blinding firewall; the on-device engine and unblinded panels are gated to authorized roles only <i>src: Identity/access management + sensitivity tags</i>	Validated + human	n/a — capability; access tied to authorized role and blinding status	Preserves trial integrity and blinding while still surfacing tier-appropriate signals
Meta-learned reference priors — Wasserstein barycenters (learning-to-learn) ADV Meta-learning the meta layer	OT barycenters + hierarchical / empirical-Bayes priors updated across studies — the meta-learning loop that sets every tier's reference. <i>src: historical cohorts/studies (per-tier sensitivity)</i>	On-device	n/a	Sets each tier's reference; recalibrated as studies complete
Nested (deep) active-inference hierarchy —	Deep / hierarchical active inference; a Wasserstein-regularized free energy ties the levels and the meta-learned	On-device	n/a	The nested loop; surfaces tier-coupled early warnings

participant → study

→ client ADV

Active inference

the architecture

priors together.

src: all tiers

Advanced engine — Optimal Transport + nested Active Inference

The rule-based KRIs answer "*has a line been crossed?*". This layer answers the earlier question: "**is this trial drifting toward a line, and where should we look next?**" It adds three things on top of the thresholds — a **geometry-aware drift** signal (Optimal Transport), an **anticipatory surprise** signal (Active Inference), and a **nested, meta-learned** structure that ties the three tiers together.

Read this first. This engine is **decision-support and advanced analytics** — it produces continuous scores and monitoring *suggestions*, never determinations. The rule-based DLT / Hy's-Law / QTcF / QTL triggers remain the **authoritative** early-warnings; OT and active inference surface risk *earlier and more continuously*, they do not replace the rules. Any model-derived signal that would inform a regulated decision must be **validated and locked** first; the validated tools still own every reported number, and the SRC / DSMB / medical monitor / PM still make every call.

1 • Optimal Transport — the geometry-aware drift signal

A threshold rule fires when one value crosses a line. **Optimal Transport** instead measures how far a whole *distribution* has moved. The **Wasserstein distance** is the minimum "cost" to transport one distribution onto another:

$$W_p(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu, \nu)} \int d(x, y)^p d\pi(x, y) \right)^{1/p}$$

— the cheapest plan π to move distribution μ (e.g. this cohort's LFTs) onto ν (the reference). The **ground metric** $d(x, y)$ encodes the outcome geometry — a Grade 1→3 move is farther than 1→2, a large exposure jump farther than a small one — and the order p tunes how heavily the large moves are weighted (W_2 penalizes them more than W_1).

Why Wasserstein and not KL-divergence? Three reasons that matter for safety:

- **It respects the outcome geometry.** Severity ordering and exposure magnitude are built into the cost — so a shift toward *worse* grades or *higher* exposure registers as a larger distance, in the right direction.
- **It counts how far the mass moves.** A few participants sliding into a dangerous tail move mass a long way in ground-metric units — W registers that, where a mean/threshold summary misses it and KL (being geometry-blind, and prone to blowing up) tells you the distributions differ but not by how far or in which direction. The dangerous part of a dose-escalation shift *is* that tail movement.
- **It is well-defined on non-overlapping support** (where $KL \rightarrow \infty$) and gives a smooth distance and a **direction** — the transport plan tells you *which* grades/exposures/participants are driving the drift, so the flag is interpretable.

The reference each tier compares against is a **Wasserstein barycenter** of prior cohorts/studies — the distribution that minimizes the average transport cost to the historical set (this is the meta-learned prior, below):

$$\bar{\mu}_{\text{ref}} = \operatorname{argmin}_{\mu} \sum_k \lambda_k W_2^2(\mu, \nu_k)$$

— a principled pooled reference built from prior cohorts ν_k , not a hand-picked baseline.

On-device feasibility: 1-D Wasserstein is a closed-form sort (cheap), and **sliced-Wasserstein** reduces the multivariate case (labs + vitals + exposure jointly) to averaged 1-D projections — light enough to run on the **local model** for participant-level data, zero egress.

Drives, on the dashboard: participant-trajectory drift, cohort dose-escalation drift, site data-distribution anomaly (central monitoring), sponsor-behavior drift.

2 · Active Inference — the anticipatory surprise signal

Treat the monitor as an agent with a **generative model** of the trial's expected trajectory. Two quantities fall out, and both are exactly what an early-warning system wants.

(a) **Variational free energy — a tractable upper bound on surprise — is an early-warning signal.** As each observation arrives, the agent updates its beliefs to minimize free energy (variational Bayesian belief-updating); the free energy **upper-bounds the surprise** (the surprisal $-\ln p(o)$) — how poorly the model explains what it just saw. The same F has two equal forms — one that shows the bound, one that you actually compute:

$$F = D_{\text{KL}}[q(s) \parallel p(s|o)] - \ln p(o) \geq -\ln p(o) \quad (\text{the bound: the KL is } \geq 0) \\ = D_{\text{KL}}[q(s) \parallel p(s)] - E_q[\ln p(o|s)] \quad (\text{complexity} - \text{accuracy} - \text{what you compute})$$

— minimizing F over beliefs $q = \text{perception}$. Under exact inference the bound is tight ($F = -\ln p(o)$ when $q = p(s|o)$); with a restricted belief family it is a generally non-tight upper bound. Either way, a participant, cohort, or program whose observations are *surprising* under the expected model drives F up *before* a frank threshold event.

The trial's **safety/quality tolerances (the QTLs, exposure caps, expected enrollment)** are encoded as the model's **preferred observations** (prior preferences C) — which enter the *expected* free energy below, not the perceptual F above. So there are two distinct early signals worth watching: **surprise** (F rising — the trial is doing something the model didn't expect) and **preference-divergence** (outcomes drifting from the preferred QTL state). They usually move together, but not always — a trial can drift toward a fully *predicted* bad state (low surprise, high preference-divergence), and you want to catch both.

(b) **Expected Free Energy = what to monitor next.** The agent chooses its next *action* — here, a monitoring/data-collection action — by minimizing **expected** free energy over candidate observations:

$$\pi^* = \operatorname{argmin}_{\pi} G(\pi), \quad G(\pi) = -E[\text{information gain}] - E_q[\ln p(o|C)]$$

— the agent **minimizes** G ; because both terms are negated, that **maximizes** expected information gain and the expected log-preference $\ln p(o|C)$ (where C is the preferred / safe outcome — the utility). The **epistemic** term picks the assessment that most reduces uncertainty about emerging risk; the **pragmatic** term favors reaching the safe, on-track state. Together they answer "which unscheduled assessment would tell us the most about whether this is a real DLT?"

So the engine doesn't just wait — it **recommends** an unscheduled LFT/ECG/PK draw, or which site/cohort to monitor next (precision-weighted attention = risk-based monitoring that focuses where information-gain is highest). **Every such action is a suggestion for human approval — never an autonomous clinical or dosing action.**

Drives, on the dashboard: participant free-energy spike + EFE-recommended monitoring; study-level free-energy drift + EFE-guided RBM; program-level free-energy.

3 · The nested / meta-learning structure

This is where the two methods compose. The three tiers are **not** three separate dashboards — they are one **deep (hierarchical) active-inference model** that minimizes a single hierarchical free energy by local message-passing. Each tier's beliefs are the **top-down priors** for the tier below; each tier's **prediction errors** (surprise) propagate **up**; and the levels run at different timescales (fast at the bedside, slow at the program). Because the levels are coupled in the model, a participant DLT propagates upward as study- and program-level risk instead of staying siloed.

META-LEARNING

Wasserstein-barycenter priors, learned across prior

CLIENT / PROGRAM

generative model of the program & sponsor trajectory

slow · weeks–months

cohorts / studies (empirical Bayes) — recalibrated as each study completes

sets & recalibrates each tier's priors ▶

OT sponsor-behavior drift

AI program free-energy

↓ top-down priors (predictions)

bottom-up prediction errors (surprise) ↑

STUDY

medium · days–weeks

dose–toxicity, enrollment & QTL beliefs

OT cohort distribution drift

AI study free-energy · EFE→RBM

↓ top-down priors (predictions)

bottom-up prediction errors (surprise) ↑

PARTICIPANT

fast · minutes–days

each participant's expected safety / PK trajectory

OT participant-trajectory drift

AI free-energy spike · EFE→next assessment

↓ observations: central-lab LFTs · ECG / QTcF · PK exposure · AEs — processed on-device (zero egress)

Meta-learning (learning-to-learn) sets the priors. Each level's generative model — expected AE rates, the dose–toxicity prior, the enrollment curve, the "preferred" QTL states — has **hyperparameters estimated by pooling across prior cohorts/studies** (an empirical-Bayes / hierarchical prior). In the hierarchy the higher level *supplies* the prior for the level below; what is *meta-learned* from the population is the **hyperprior** that shapes those priors. Concretely, the **Wasserstein barycenter** of past cohorts is used as the reference a new cohort is scored against, and it **recalibrates as each study completes** — so a new study starts population-informed instead of from scratch. That cross-study hyperprior is the "meta".

The tie that binds: a Wasserstein-regularized objective. Using an Optimal-Transport cost in place of (or alongside) the KL term in the variational objective makes the divergence **geometry-aware and well-defined on non-overlapping support** — the same property that made W the right drift metric now makes the belief-updating robust, and it lets the meta-learned barycenter prior enter directly. The honest trade-off: once you swap KL for W it is **no longer the strict variational free energy**, so the clean "upper bound on surprisal" guarantee above relaxes into a geometry-aware regularizer — a deliberate, research-grade design choice, not a free lunch. OT supplies the *metric*; active inference supplies the *dynamics*; meta-learning supplies the *priors*.

What this buys you over thresholds alone. (1) **Earlier** — drift and surprise rise before a line is crossed. (2) **Geometry- and tail-aware** — it catches mass moving toward danger, not just a single value. (3) **Anticipatory** — it proposes the most informative next assessment instead of waiting. (4) **Coherent across tiers** — one nested model, so participant-level risk correctly propagates to study and program risk, with priors that get smarter every study.

Honesty & validation

- **Advanced ≠ validated.** These are research-grade methods. Treat their outputs as *triage and prioritization* until the specific implementation is validated and version-locked; the rule-based triggers and the validated safety/dose-escalation tools remain authoritative for any regulated decision.
- **Frozen for reproducibility.** If a model-derived signal ever feeds a submission or a stopping decision, the model, the reference barycenters, and the thresholds must be **frozen and documented** (the same model-freeze argument as the rest of the program) so a result is reproducible.
- **Participant-level stays on-device.** The participant-tier OT and active-inference computation runs on the local model with zero egress; only de-identified aggregates inform the cloud tiers.
- **The action is a suggestion.** EFE recommends *what to observe*; a human orders it. Nothing about dosing, escalation, or termination is automated.

Governance — the AI flags, humans decide

- **The AI never makes a safety or termination call.** It detects, aggregates, triages, explains, and drafts the alert. The medical monitor / SRC / DSMB make DLT & safety determinations; the PM owns the study; the account lead owns the client signal.
- **Participant-level safety / PK / unblinded data never leaves the device.** The local LLM runs on-prem / on-device with zero egress. Only non-sensitive operational & commercial signals use cloud Copilot (under BAA).
- **Validated tools own the numbers.** Reported safety numbers come from the validated safety database; cohort DLT-rate / stopping-rule math comes from the validated dose-escalation tool — never an LLM.
- **Early-warning, not auto-action.** A flag triggers a human review and a disposition, logged with a 21 CFR Part 11 audit trail (ALCOA++). Nothing is paused, escalated, or terminated by the system itself.
- **The client tier is decision-support for the account team** — surfacing commercial signals, never a unilateral judgement about a sponsor.

Standing it up

Phase 1 — the safe wins (cloud)

- Study & client tiers first: wire enrollment / KRI / QTL feeds and the finance-AR / CRM / sponsor-comms signals into cloud Copilot; build the Study and Client RAG panels behind a human-review queue. De-identified / non-sensitive only.

Phase 2 — the on-device participant tier

- Stand up the on-device local LLM in the validated environment; connect the safety DB / central-lab / ECG feeds; encode the DLT criteria, Hy's-Law / eDISH logic, QTcF and PK-exposure thresholds; route every participant-level signal on-device with zero egress.
- Wire the alerting to the medical monitor / SRC and the disposition log.

Phase 3 — scoring, escalation & audit

- The weighted roll-up to tier RAG, the escalation routing, the Part-11 audit trail and human-acknowledgement, and the cross-study safety view for the same compound.

Worked example — a DLT early-warning fires

Day 9, cohort 4 (dose escalation)

Participant **S-1042**'s central-lab feed posts: **ALT 3.4× ULN, total bilirubin 2.1× ULN, ALP 1.3× ULN**. The **on-device** local LLM evaluates the eDISH logic, recognizes a **Hy's-Law pattern**, and — because it's inside the DLT window for this dose — flags a **candidate DLT**. Nothing left the device.

Within minutes

The model pre-assembles the case: the LFT trajectory, the eDISH context, the participant's dose & exposure, concomitant meds, and the running **cohort DLT tally vs the 3+3 rule**. It drafts the alert and routes it to the **medical monitor and the SRC chair** — *ahead* of the scheduled cohort review — and logs the flag with a Part-11 audit entry.

The humans decide

The medical monitor orders a confirmatory repeat LFT; the SRC convenes. Per protocol, further dosing/escalation pauses pending adjudication. The **validated safety database** holds the authoritative numbers; the SRC makes the DLT determination. The dashboard's Participant panel goes **red**, the Study panel flags the cohort, and a cross-study check runs for the same compound.

The payoff

The signal surfaced on **day 9**, not at the cohort meeting on day 21 — with the SRC packet already built, the data never leaving the device, and a human making every call.

FAQ

Does the AI stop a study or a dose? No. It is an *early-warning* system — it flags, aggregates, and drafts; the SRC / DSMB / medical monitor and the PM make every decision.

Why on-device for the participant tier? It's participant-level safety / PK / unblinded data. On-device with zero egress is the only way to use AI on it without a data-residency problem.

Where do the numbers come from? Validated systems — the safety database and the validated dose-escalation tool. The LLM never produces a reported safety number or a DLT-rate verdict.

Is the client tier appropriate? It's commercial decision-support for the account team — surfacing AR, communication, scope, and sponsor-health signals — not a judgement the system makes on its own.

How does it relate to the rest of the program? It's the risk-monitoring layer of the hybrid AI strategy: on-device local for sensitive data, M365 Copilot for non-sensitive — the same routing principle, pointed at early-termination risk.

What are the "ADV" / Optimal-Transport / Active-Inference signals? An [advanced analytics layer](#) on top of the thresholds: **Optimal Transport (Wasserstein)** measures how far a distribution has drifted from a meta-learned reference (earlier and more tail-sensitive than a single value crossing a line), and **nested Active Inference** treats the monitor as a generative model whose *surprise* is an early-warning and whose *expected free energy* recommends the most informative next assessment. They are decision-support — advanced, not yet validated — and never override the rule-based triggers or the humans.

Walkthrough transcript

The complete narration of the walkthrough video, as readable text.

1. Part 1

This is a short walkthrough of the trial-risk early-warning dashboard. The idea is simple: catch the risk that a trial might have to stop — early, and across three levels: the individual participant, the study as a whole, and the client relationship. It runs on a hybrid of Microsoft Copilot and on-device local models. And throughout, the AI only detects, aggregates, and triages. It never makes the safety or termination call. Let me show you how it works.

2. Part 2

Early-phase trials can stop for reasons that build up quietly. A participant's safety signal. A cohort's toxicity creeping toward the stopping rule. Or a sponsor going cold on the program. The trouble is that the first formal look is often the cohort review or a governance meeting — and by then we are days or weeks behind. The dashboard watches all three of these continuously, so the earliest mover is visible long before that meeting.

3. Part 3

Here is the dashboard itself. Three panels — participant, study, and client — each with a simple red, amber, green status, the top signals underneath, and the engine that powers it. Below the panels is a live early-warning feed. Right now everything is amber: a candidate liver signal on one participant, an enrollment limit breached at the study level, and a payment-aging flag on the client side. Each panel rolls its signals up to a status — but that status is a prompt for a human, not a verdict the system reaches on its own.

4. Part 4

The participant tier is the most sensitive, so it runs entirely on-device, with zero egress. It flags candidate dose-limiting toxicities inside the evaluation window: Hy's-Law liver signals, QTc prolongation, PK exposures beyond the safety margin, and escalating adverse-event clusters. The key word is early. It surfaces the pattern and pre-assembles the safety-committee packet before the formal cohort review. What it does not do is adjudicate the toxicity. That is the medical monitor and the Safety Review Committee.

5. Part 5

The study tier watches the trial's trajectory against its own rules: the cohort toxicity rate against the stopping rule, enrollment and screen-failure limits, the safety quality-tolerance limits from ICH, and central data-quality signals. These are de-identified, aggregate numbers, so they run on cloud Copilot — with any participant-level drill-down dropping back to the on-device model. The project manager owns these calls, with the safety committee on anything safety-related.

6. Part 6

The third tier is the commercial one — the early warning that a sponsor might cancel, de-scope, or not renew. Payment aging, a drop in communication, scope-reduction change orders, and the sponsor's own financial and pipeline health. This is non-sensitive business data, so it sits on cloud Copilot. And I want to be clear about what this

tier is. It is decision-support for the account team, surfacing signals. It is never a judgment the system makes about a sponsor.

7. Part 7

Underneath, one rule decides where everything runs: sensitivity. Participant-level safety and unblinded data go to the on-device model, and never leave the validated environment. Non-sensitive, aggregate operational and commercial data go to cloud Copilot, under a business-associate agreement. And here is the rule that governs the whole thing. The AI flags, aggregates, triages, and drafts the alert — it never makes the safety or termination call. The reported numbers come from the validated safety database, never the model. Every flag triggers a human review and a logged disposition, and nothing is paused or terminated by the system itself.

8. Part 8

There is an optional advanced layer for teams that want it. Instead of waiting for a threshold to be crossed, it watches for drift and surprise. Optimal transport measures how far a distribution has moved from a learned reference, in a way that respects clinical severity. Active inference treats the monitor as a model of the expected trajectory, so rising surprise is itself an early warning — and it can even recommend the most informative next test to run. It is decision-support only: the threshold rules stay authoritative, the methods were independently checked, and it is validated and frozen before it ever informs a regulated decision.

9. Part 9

Here is how it comes together. Day nine of cohort four: a participant's labs post a Hy's-Law pattern. The on-device model recognizes it, and because it is inside the evaluation window, flags a candidate toxicity — without any data leaving the device. Within minutes it assembles the case and routes it to the medical monitor and the committee chair, ahead of the scheduled review. The humans take it from there: a confirmatory test, the committee convenes, dosing pauses per protocol. The payoff is the timing. This surfaced on day nine, not at the cohort meeting on day twenty-one — with the packet already built, and a human making every call.

10. Part 10

Building it is staged. First, the safe wins: wire the study and client tiers from existing operational and commercial feeds into cloud Copilot, behind a human-review queue, with de-identified data only. Second, the on-device participant tier in the validated environment, with the safety logic encoded and alerting wired to the medical monitor. Third, the scoring, escalation, and audit trail — and, optionally, the advanced engine, validated and frozen before it informs anything. Each phase delivers value on its own.

11. Part 11

So, in one line: catch termination risk before the meeting, and let humans make every call. It is earlier — a liver signal on day nine, not day twenty-one. It is three tiers on one screen. It is governed by design — sensitive data on-device, validated tools owning the numbers, and the committees and project managers deciding. And it fits the program: a phase-one pilot deliverable on the same hybrid stack — no new principle, just a new target. Thanks for watching.